



DEPARTMENT OF COMPUTER SCIENCE

IT3212 - DATA POWERED SOFTWARE

Assignment 1 - Data Preprocessing

Group 23

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10.2025

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1 Introduction

The Smoking and Drinking Dataset originates from the Korean National Health Insurance Service and contains health examination data collected from a large sample of the population, presumably in Korea. It was constructed to analyze different kinds of body signals, and to classify a person as a smoker or drinker, and their correlation with several physiological indicators. The dataset includes a wide range of features, which can be seen in the table 1. The combination of physiological markers and whether a person smokes or drinks enable studying correlations between health and lifestyle. [1]

We thought an appropriate use for this data would be to classify *healthy* persons (i.e. a general representation of the average person, no abnormal clinical values which could indicate acute disease etc.) as a drinker or not, and will proceed with preprocessing the data with that in mind. We will perform an initial exploratory analysis of the data, clean it so it better represents healthy people, do transformations and splitting into training and testing sets, and finally we will perform dimensionality reduction.

Feature	Description	Unit
sex	Male, female	na.
age	Age of person (rounded to the nearest five)	years
height	Height of person (rounded to the nearest five)	cm
weight	Weight of person (rounded to the nearest five)	kg
waistline	Circumference around the waist	cm
sight_left	Decimal visual acuity (=1) normal, (>1) better, (<1) worse [2]	na.
sight_right	Decimal visual acuity (=1) normal, (>1) better, (<1) worse [2]	na.
hear_left	Normal (1) or abnormal (2) hearing in left ear	na.
hear_right	Normal (1) or abnormal (2) hearing in right ear	na.
SBP	Systolic blood pressure	mmHg
DBP	Diastolic blood pressure	mmHg
BLDS	Blood sugar (glucose) level after fasting	mg/dL
tot_chole	Total cholesterol	mg/dL
HDL_chole	High-density lipoprotein cholesterol	mg/dL
LDL_chole	Low-density lipoprotein cholesterol	mg/dL
triglyceride	Triglyceride level	mg/dL
hemoglobin	Hemoglobin level	g/dL
urine_protein	Protein in urine: 1(-), 2(+/-), 3(+1), 4(+2), 5(+3), 6(+4)	na.
serum_creatinine	Blood (serum) creatinine	mg/dL
SGOT_AST	Glutamate-oxaloacetate transaminase (Aspartate transaminase)	IU/L
SGOT_ALT	Glutamate-oxaloacetate transaminase (Alanine transaminase)	IU/L
gamma_GTP	γ -glutamyl transpeptidase	IU/L
SMK_stat_type_cd	Capacity of smoking: 1 (never), 2 (quit), 3 (still)	na.
DRK_YN	Whether the person drinks or not	na.
BMI	Added manually calculated using height and weight [3]	na.

Table 1: Overview of dataset

2 Data Exploration

The data exploration phase started with printing the key characteristics of the dataset, like the number of entries and categories, as well as the names of the columns. From here we printed out the 10 first rows of the data to get an overview of how the data looked.

To get a further look into the distribution of the various health values for each individual in the dataset, we used the Pandas method `describe()` to get count, mean, standard deviation, maximum, and minimum, as well as the 25th, 50th, and 75th quartile.

We also printed out the data type of each column to more clearly identify numerical and categorical features.

From here, we identified the number of missing values for each column, which turned out to be 0, meaning that there is no missing data in the dataset, at least not represented as NaN values.

To get a deeper understanding of the categorical features we displayed the unique values and their value counts.

Furthermore, we displayed box plots of the column values in order to visualize the distribution of the data, and attempt to identify outliers visually, however we found that histograms and Q-Q plots was better for this.

3 Data Cleaning

Our dataset contains no missing values, however we would still like to consider how missing data would be handled if it was present. Missing data can come from many different sources, for example mistakes in data collection, for instance if the data was collected with surveys and some data were not provided. Another source of missing data could be system malfunctions. [4]

An approach to handling missing data is deletion, where rows or columns containing missing values are removed. This method is simple, but it can lead to losing a significant amount of information, especially if a large portion of the dataset is affected. However, this can be a good strategy when large parts of rows and/or columns are affected. [4]

Another common strategy is imputation, where missing values are replaced with estimated ones. For example, numerical features can be imputed with the mean or median, whichever fits the specific data better. More advanced approaches to handling missing values include linear and polynomial interpolation, k-nearest neighbors, or model-based methods, like regression imputation. Another trick would be to add some random noise to the imputed data to simulate reality and restore normality. [4]

4 Handling Outliers

We tried using the interquartile range (IQR) method. to detect outliers, but this resulted in a very high percentage of the data being marked as outliers. If we were to just delete these rows it would lead to severe data loss and bias. After discussing other options we concluded that we would manually detect abnormal values in each feature, based on abnormal health metrics.

We plotted each feature using histogram & KDE and Q-Q plots. This way we could more easily see the distributions of each feature and detect actual outliers.

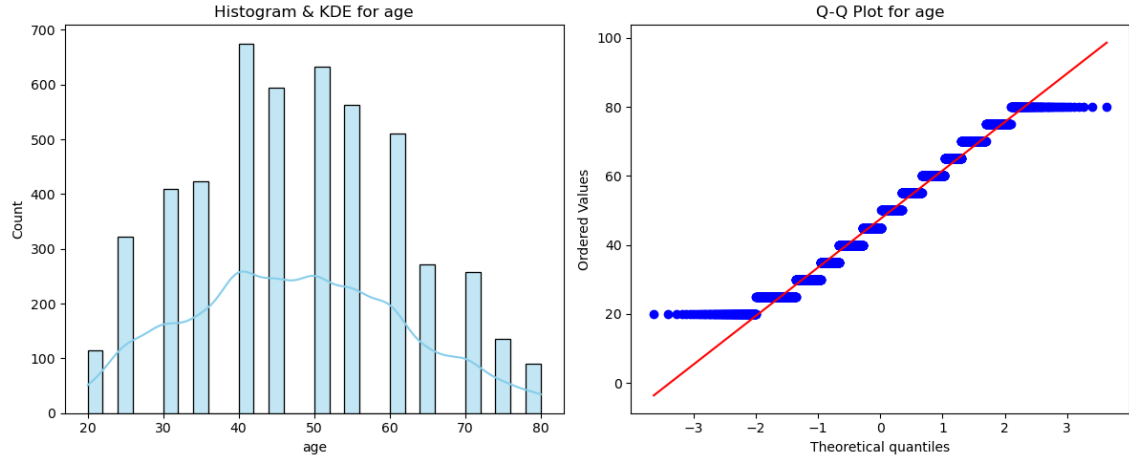


Figure 1: Histogram and Q-Q plot for age

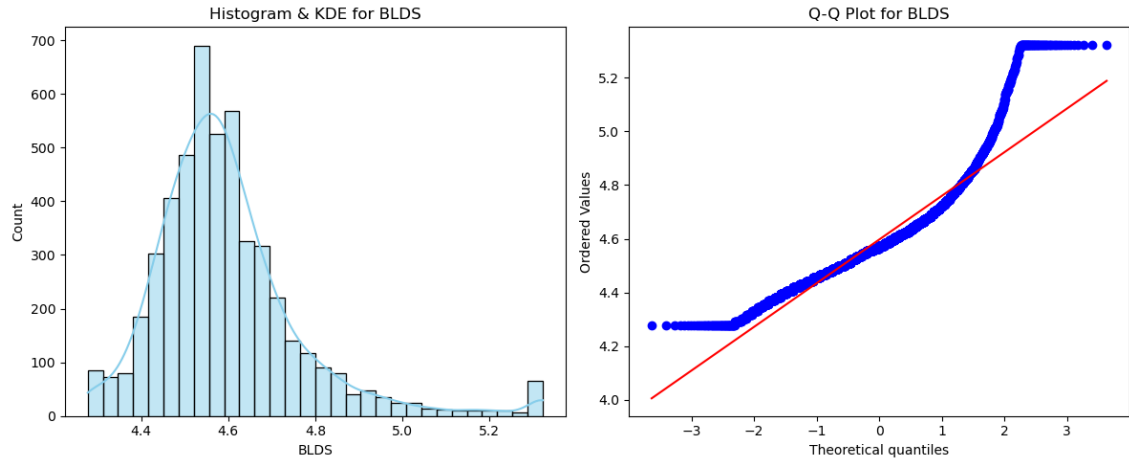


Figure 2: Histogram and Q-Q plot for BLDS

Justification for handling outliers in the different features:

- **age:** No sign of outliers, considering that the age is rounded to the nearest five.
- **height:** We use BMI to determine if the height is abnormal or not.
- **weight:** We use BMI to determine if the weight is abnormal or not.
- **BMI:** Based on BMI, you are considered extremely underweight and are therefore unhealthy when you have less than 16, and extremely overweight when your BMI is greater than 40, so we remove these values. [5]
- **waistline:** We set the upper limit for waistline at 150. 145 is the biggest waistline in the dataset that is not 999 (seemingly a placeholder for missing or not measurable). The smallest recorded waistline in the world was 38,1 cm. Therefore we set the lower limit at 38,1. [6]
- **sight_left:** Datapoints recorded as 9.9 are removed.
- **sight_right:** Datapoints recorded as 9.9 are removed.
- **SBP:** We removed values over 150 and values below 90 as these are considered abnormal. [7]
- **DBP:** We removed values over 90 and values below 60 as these are considered abnormal. [7]

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- **BLDS:** We removed values under 60 and above 126 (126 and above indicate diabetes).
 - **tot_chole:** For cholesterol there are a lot of entries with high cholesterol. According to [8] 24% of the Korean population have high cholesterol (having above 240 mg/dL) in 2022, which matches well with our observation of 24% of the data having high cholesterol. We still choose to remove at above 1000 since these values are extremely abnormal and signify an acute disease.
 - **HDL_chole:** Removed 1 datapoint that we deemed unrealistic.
 - **LDL_chole:** Removed values over 2000.
 - **triglyceride:** We removed values above 500 since these are considered severe amounts [9].
 - **hemoglobin:** We remove hemoglobin at over 20 and under 5 as these levels are dangerous. Hemoglobin levels below 5 can cause heart failure and above 20 can lead to blood clots. [10]
 - **serum_creatinine:** Serum creatinine values bigger than 4 mg/dL indicate seriously impaired renal function [11] [12]. Since there are a notable amount of values bigger than that, we removed all values exceeding 15 mg/dL from the dataset. This to reduce the chance of not representing either critical clinical conditions, but at the same time avoiding data entry errors.
 - **SGOT_AST:** Remove at above 1000 to remove placeholders.
 - **SGOT_ALT:** Remove at above 1000.
 - **gamma_GTP:** Remove datapoints with values at 999.

After removing these outliers we have 890 077 data points left, this is 89.78% of the original data points.

It is important to note that none of the members of our team are medical professionals. While we have made efforts to incorporate domain knowledge when handling outliers in the dataset, our decisions are based on publicly available information and general statistical reasoning. As such, there is a possibility that some of our assessments may be inaccurate or incomplete.

Medical datasets often contain extreme values due to the nature of health conditions and diseases, which can significantly affect biological measurements. The values may reflect underlying medical issues rather than data errors. Therefore, we have approached outlier detection and treatment with caution, in order to preserve medically relevant variation while minimizing statistical distortion.

5 Data Transformation

Encoding Categorical Data

Encoding categorical data in numerical form is an important step in preparing a dataset for data science and machine learning purposes. Since most relevant algorithms rely on mathematical methods that require numerical values, encoding ensures that categorical features contribute meaningfully.

For this dataset, the only categorical values that were not already encoded are **sex** and **DRK_YN** (Drinker or Not). **SMK_stat_type_cd** (smoking state), **hear_left**, and **hear_right** are also categorical data, but was already encoded into numerical values.

For encoding **DRK_YN**, we chose to set 'N' to 0, and 'Y' to 1. Similarly, for **sex** we set 'Male' to 0, and 'Female' to 1. Since they are binary values in both cases, this was deemed the most fitting and obvious solution. We deemed that one-hot encoding would not be preferable due to the way the dataset is structured, as well as the other categorical values being label encoded.

Feature Scaling

Feature scaling is an important part of making sure that you have effective features to improve model performance and reliability of machine learning algorithms [13] [14]. Effective features help the model capture important patterns and relationships in the data. Following are key advantages of feature scaling:

- **Improves model performance:** Scaling ensures that features are presented on comparable scales, which helps models learn more effectively and often leads to better accuracy and predictive power [13] [14].
- **Faster convergence during training:** Helps gradient-based algorithms train faster and more reliably. Feature scaling helps gradients behave consistently across dimensions, allowing the model to speed up convergence [13] [14].
- **Balanced feature influence:** Without scaling, features with larger numeric ranges can dominate the learning process, overshadowing smaller-scale features. By transforming the data to a standardized range, we ensure that all features contribute fairly to the model's decisions [13] [14].
- **Increases numerical stability:** Scaling reduces the risk of numerical issues such as overflow or underflow during computation [14].
- **Facilitates algorithm compatibility:** Certain machine learning models, particularly those based on distance metrics and gradients, assume that features are on similar scales. Without feature scaling, the distance calculations could be skewed, leading to biases in the model and potentially misleading results. Feature scaling normalizes the range of features so that each one contributes equally to the distance calculations [13] [14].

Justification of feature scaling technique

For our dataset, we chose to apply *standardization (Z-score normalization)* as our feature scaling technique. This decision was based on both the nature of the data and the pre-processing steps we performed.

Medical data often contains features with different units and scales, such as *waistline* (cm), *triglyceride* (mg/dL), and *gamma_GTP* (IU/L), which can vary significantly in magnitude. Standardization transforms each feature to have a mean of 0 and a standard deviation of 1, ensuring that all features contribute equally to the learning process [14] [15].

Another reason we chose Z-score standardization is that it's less sensitive to outliers than *Min-Max scaling*, which was another technique we [14] [15]. Even though we have performed outlier detection and removal prior to scaling, extreme values in medical data are not unusual and can be clinically meaningful.

Standardization is particularly well-suited for our use case because:

- It preserves the distribution shape of each feature, which is important when working with medical data.
- It is robust to moderate outliers [15].
- After analyzing histograms and Q-Q plots of our dataset, we observed that many features appear approximately normally distributed, while others exhibit varying degrees of skewness. We chose to apply Z-score standardization, which centers each feature around a mean of 0 and scales it to unit variance. Although this transformation does not fully normalize skewed distributions, it reduces the impact of scale differences and brings features closer to a comparable range. This is particularly beneficial for models that assume normally distributed inputs.

It is important to note that we have not selected a specific ML-model for our analysis. Different models may benefit from different types of scaling. For example, distance-based models such as k-NN and SVM are highly sensitive to feature magnitude and often perform best with Min-Max scaling, while models that assume normally distributed inputs, such as logistic regression or linear models, may benefit more from standardization.

Our choice of feature scaling technique was made independently of any particular ML-algorithm, and was based on the nature of the dataset. It may have been different if choosing ML-model was a part of the assignment.

6 Data splitting

Splitting the data set into training and validation sets is a crucial step to ensure that a model can generalize well to unseen data [16]. Without this separation, the model may learn patterns specific to the training data (overfit) and perform poorly on new data.

A common strategy is to first split the data into a training set and a test set with a 80/20 split. The training set is then used for model development and validation, while the test set is held out for final evaluation. This ensures that the model is not influenced by the test data during training. To further reduce the risk of overfitting, we would use *cross-validation (CV)* to validate the model within the training set [17, pp. 644–645]. CV is a model evaluation strategy that involves dividing the available data into subsets to train and test the model multiple times.

More specifically we would use the method *k-fold cross-validation*. In k-fold CV, the training data is divided into k equal parts. The model is trained k times, each time using $k - 1$ folds for training and one fold for validation. This ensures that every data point is used for both training and validation. The final performance is aggregated for all k iterations, providing a reliable estimate of the model's generalization ability.

By first splitting the data and then applying CV, the risk of overfitting is reduced since the model is evaluated on distinct subsets of the data.

7 Bonus Task

PCA works by combining highly correlated features into principal components. This can be good for efficiency, reducing redundancy and model training. PCA was applied after feature scaling since it is sensitive to feature magnitudes.

By applying PCA we reduce the amount of features while still being able to explain 95% of the variance. This way we went from 22 features to 17 principal components.

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